



Multiplicative relationships and direct proportion

Task A: Direct proportion

1) Which of the following show a **directly proportional** relationship between A and B?

a)

A	B
3	12
6	24
8	32
10	40

b)

A	B
4	7
6	11
8	15
10	19

c)

A	B
5	35
15	105
20	140
30	210

d)

A	B
4	2
7	3.5
9	4.5
14	7

e)

A	B
3	13
4	16
5	19
8	28

f)

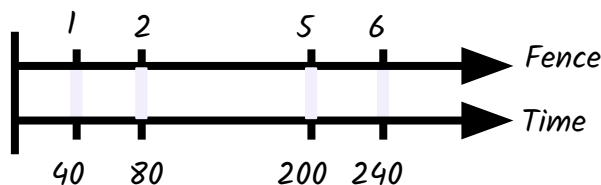
A	B
100	25
80	20
75	18.75
40	10

The tables which show a directly proportional relationship are a, c, d, f.

2) Below are some variables which are **directly proportional** to each other.

a) Jun is painting his neighbour's fence. He paints one fence every 40 minutes.

i) A double number line and mark on at least 3 equivalents

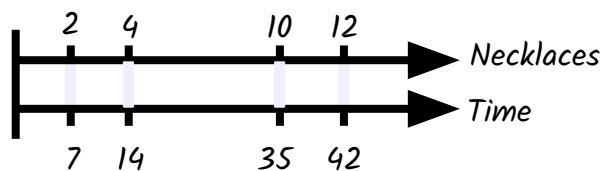


ii) A ratio table

Fence	Time
1	40
6	240

b) Andeep is making necklaces for the school fete. He makes 2 necklaces every 7 minutes.

i) A double number line and mark on at least 3 equivalents

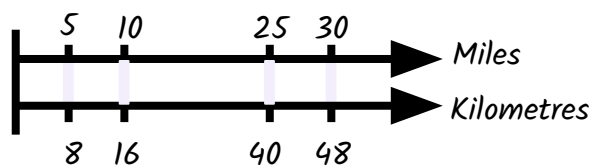


ii) A ratio table

Necklaces	Time
2	7
10	35

c) 5 miles is equivalent to 8 km.

i) A double number line and mark on at least 3 equivalents

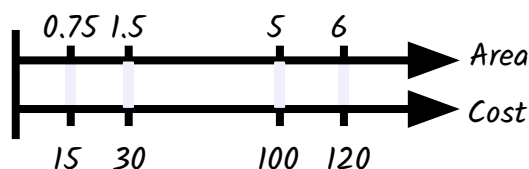


ii) A ratio table

Miles	Kilometres
5	8
10	16

d) A box of tiles covering 0.75 m^2 , costs £15

i) A double number line and mark on at least 3 equivalents

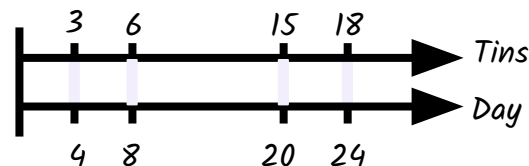


ii) A ratio table

Area	Cost (£)
0.75	15
10	200

e) Sofia's dog eats 3 tins of food every 4 days.

i) A double number line and mark on at least 3 equivalents



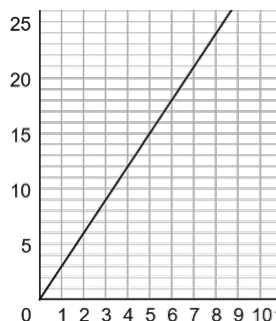
ii) A ratio table

Tins	Day
3	4
15	20

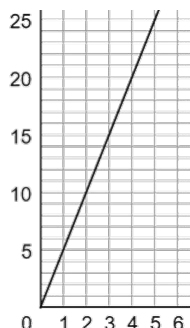
Task B: Graphing proportional relationships

1) Match each table to the correct graph and fill in the missing values:

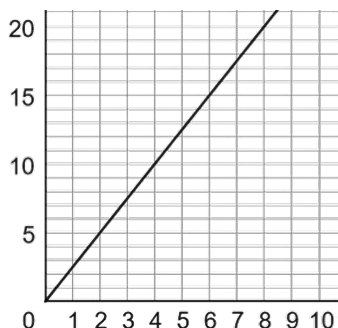
A



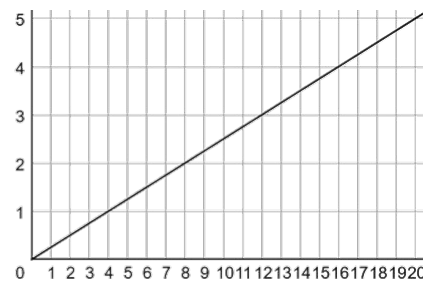
B



C



D



2	5
7	17.5
3	7.5
8	20

C

2	6
5	15
3	9
4	12

A

4	1
6	1.5
8	2
16	4

D

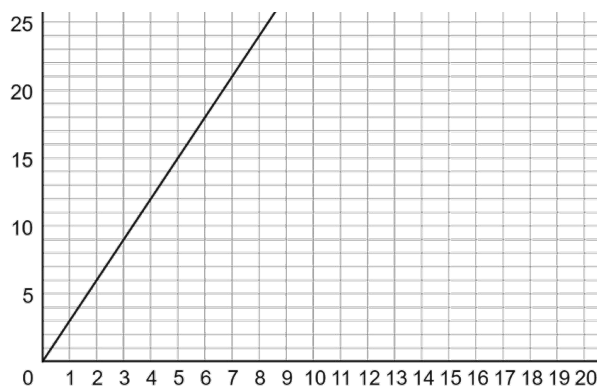
3	15
7	35
2	10
5	25

B

2) The following all have a **directly proportional** relationship. Plot a graph of each.

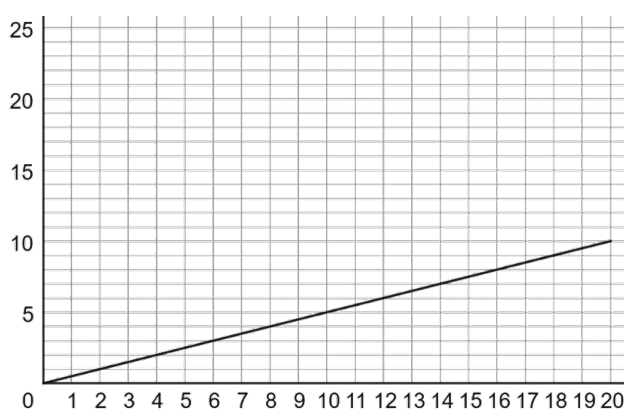
a)

2	6
3	9
4	12
6	18



b)

4	2
8	4
12	6
16	8



c)

1	1.5
2	3
4	6
8	12



3) Match each double number line to the correct graph:

